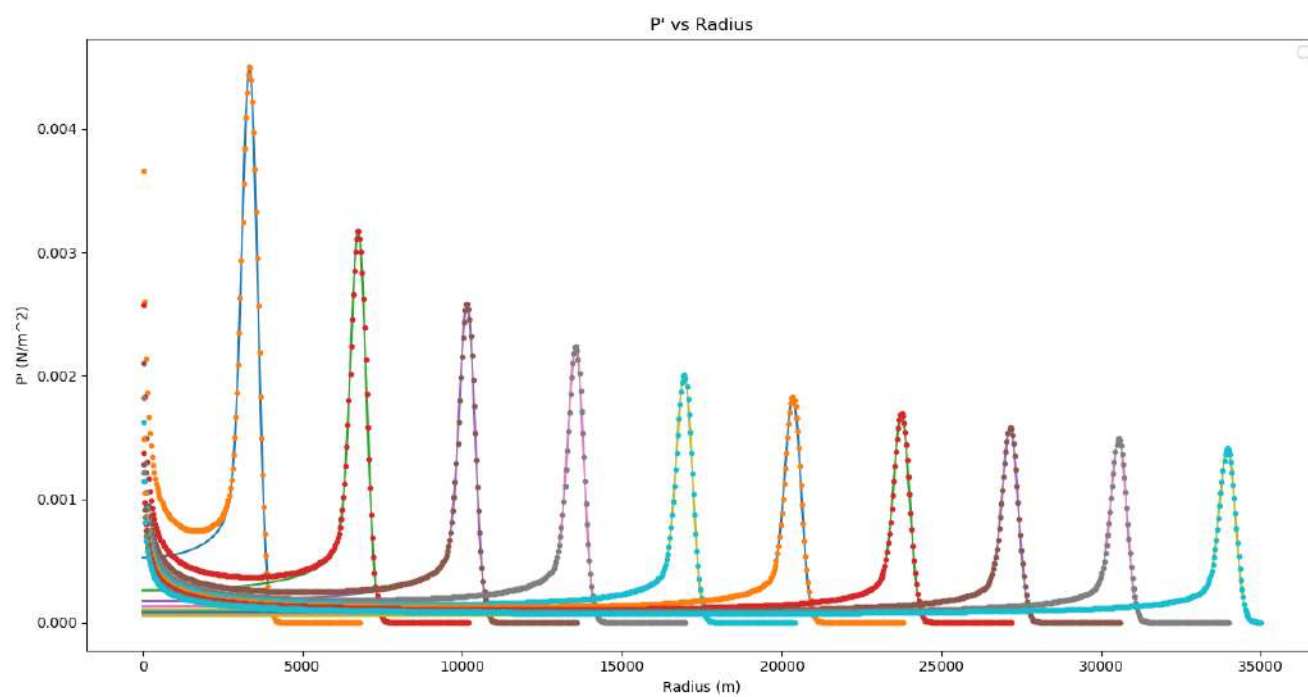
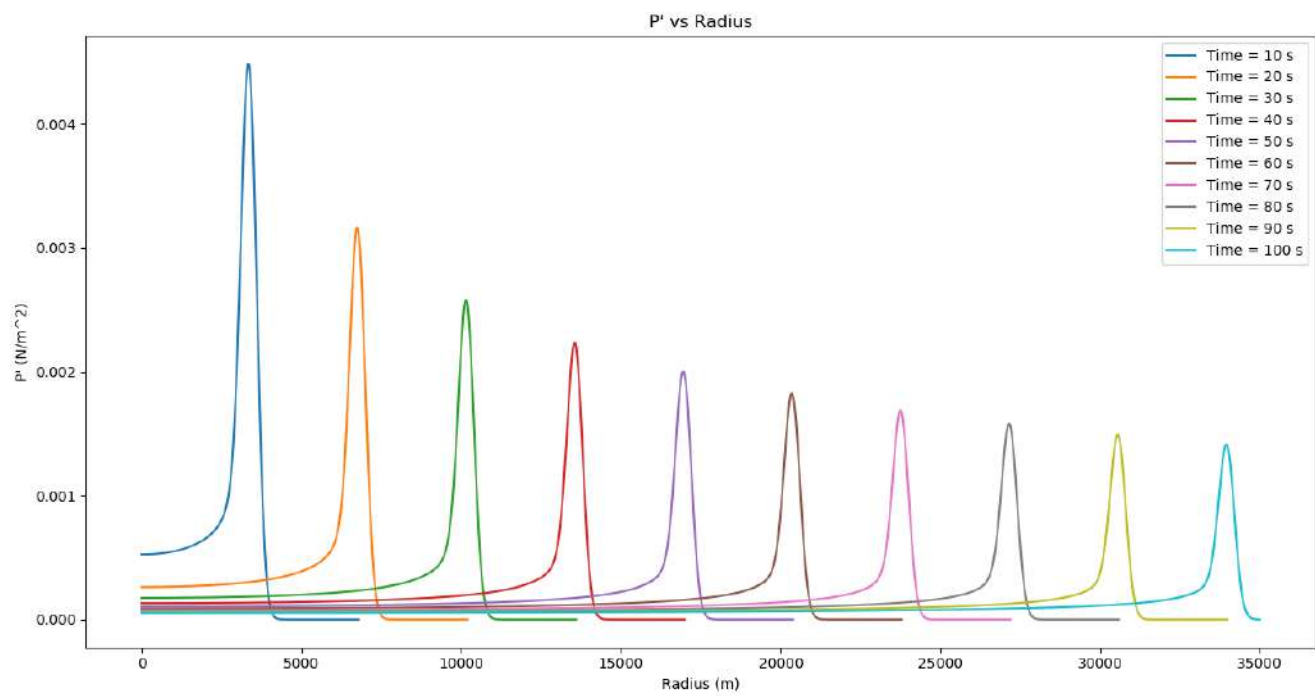
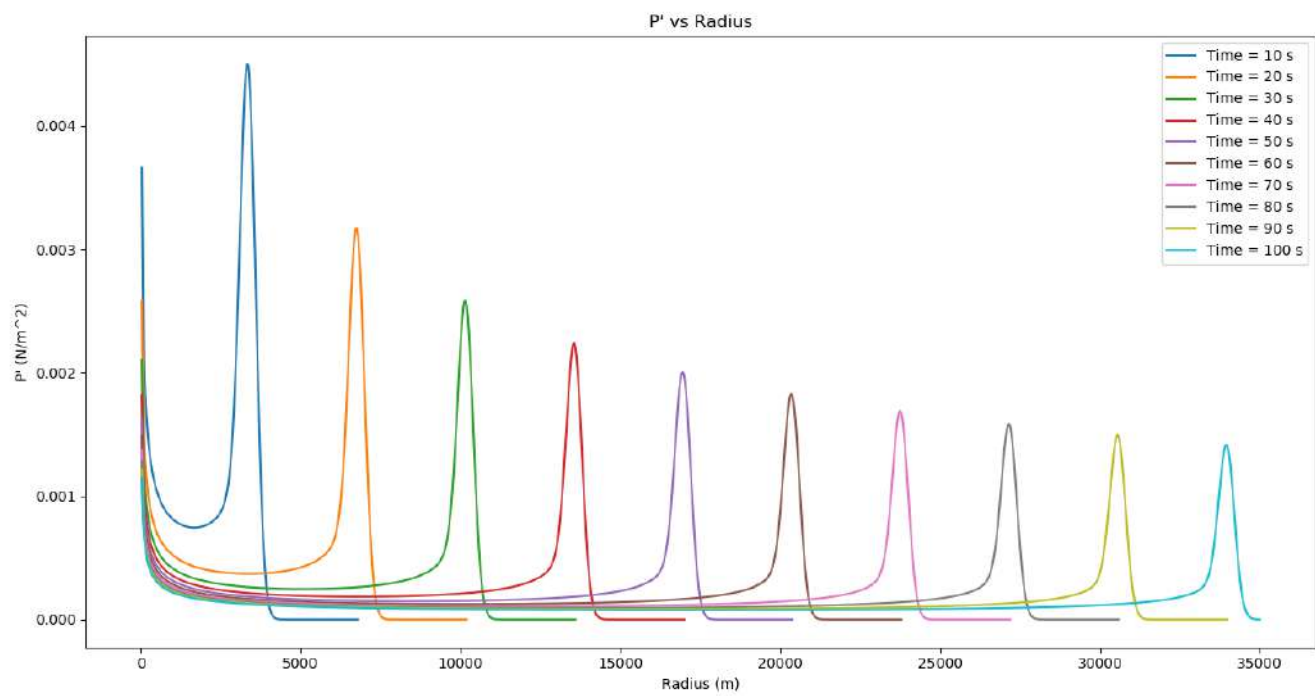








```

import numpy as np
import matplotlib.pyplot as plt

c=340.0
def P_prime(tau,c,t,r):
    dtau=tau[1]-tau[0]
    F=np.exp(-tau**2) / np.sqrt(c**2 * (t-tau)**2 - r**2)
    return (F[1:]+F[:-1])*0.5*dtau

def P_prime_far(tau,c,t,r):
    dtau=tau[1]-tau[0]
    F=(np.exp(-tau**2) / np.sqrt(t-r/c-tau)) *
(1/np.sqrt(2*r*c))
    return (F[1:]+F[:-1])*0.5*dtau

def integrate(c,t,r,func):
    tau=np.linspace(-10,t-r/c-0.000001,1000)
    return np.sum(func(tau,c,t,r))

def plot1():
    p_prime=[[ ] for i in range(10)]
    p_prime_far=[[ ] for i in range(10)]
    iter_1=0
    plt.figure(1)
    for t in range(10,110,10):
        k=np.linspace(0.0,35000,1000)
        for r in k:
            p_prime[iter_1].append(integrate(c,t,r,P_prime))

p_prime_far[iter_1].append(integrate(c,t,r,P_prime_far))
        plt.plot(k,p_prime[iter_1])
        plt.plot(k,p_prime_far[iter_1],'.')
        iter_1 += 1
    plt.title("P' vs Radius")
    plt.ylabel("P' (N/m^2)")
    plt.xlabel("Radius (m)")
    plt.legend(loc='best')
    plt.show()
plot1()

```